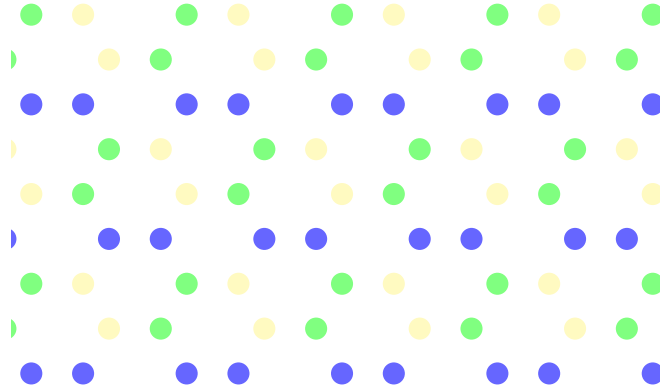


# Homework 3

## *Crystal Structures and Reciprocal Lattice*

1. Consider the following ternary, two dimensional crystal structure which repeats infinitely.



The picture is drawn to scale. Find the primitive lattice and the basis.

2. Consider NaCl. First, define the structure (use primitive unit cell). Second, determine the corresponding reciprocal lattice vectors.
3. Consider Graphene. First, define the structure (use primitive unit cell). Second, determine the corresponding reciprocal lattice vectors. Third, sketch the first Brillouin Zone.